MILITARY SPECIFICATION

' INTERVALOMETER, ORDNANCE, RELEASE TD-239/A "

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 **The equipment covered by this** Specification shall generate pulses, recurring at precise intervals of time, which are suitable for direct operation of release solenoids of bomb racks and other store suspension equipment. This equipment shall be such that the use of tables, or "graphs, is unnecessary to provide the proper time interval associated with a desired ground spacing between dropped stores when the aircraft ground speed is manually inserted. The intervalometer shall also control the number of stores expended.

2. " APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL- C- 172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-C-3098	Crystal Units, Quartz
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-5422	Enviromental, Testing, Aircraft Electronic Equipment
MIL-1-6181	Interference Limits; Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristics of

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MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment"

Bureau of Aeronautics

SAR-300 "	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

Military

MS- 25245	Connections - Input Power Airborne Electronic Equipment 28-Volt DC, 115-Volt Single- Phase AC, and/or 115/200-Volt Three-Phase AC, or Combinations

DRAWINGS

Bureau of Aeronautics

58A163D1	TD-239/A Intervalometer, Ordnance Release General Arrangement
58A163D2	TD- 239/A Intervalometer, Ordnance Release Installation
58A163J3	TD-2391A Intervalometer, Ordnance Release Schematic Wiring

2.2 Availability of Documents:

(1) Give both the title and number when requesting specifications, standards, drawings, and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (AD), 700 Robbins Avenue, Philadelphia 11, Pennsylvania.

(3) Copies of applicable drawings may be obtained upon application to the. Bureau of "Aeronautics, " Navy Department, Washington 25,D. C., Attention: Avionics Division.

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2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have Precedence over all " applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the Bureau of Aeronautics.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts" and Materials - In the selection of parts and materials, fulfillment of major design objectives -shall be- the prime consideration. In so doing the following shall govern:

(1) parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) -When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a nonstandard part is used, the meeting of general material requirements such as noninflammability, fungus resistance, nonintoxic, etc. , and the meeting of environmental conditions as required by Specifications MIL-E-5400 or MIL-E-5422 shall be of prime consideration.

(4) In the selection of parts, . the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is to be built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will . pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Nonstandard parts of an electrical nature as defined in Specification MIL-E-5400, must be approved before the preproduction model of the equipment is submitted for tests, The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Nonstandard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

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3. 1,2 Tube Types - The choice and application of electron tubes used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above shall - " . " be approved by the Bureau of Aeronautics; however, before consider- - ation for. approval, the "contractor shall submit the following information

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualificatiion approval for the particular tube.

3. 1.2.1 Electron Tube Complement - A tube complement report shall be prepared . in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes-and transistors) used in the equipment.

3.1.2.2 Crystal Units - The choice and application of crystal units used in the equipment shall be in accordance with Specification MIL-C-3098. The use of any crystal not meeting the requirements of MIL-C-3098 must be approved by the Bureau of Aeronautics and information similar to that required by 3. 1.2 (2) (a), (b), and (c) shall be furnished.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an Improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

3. 2 Design and Construction

3. 2.1 General - This equipment shall fulfill all applicable design and construction requirements of Specification MIL-E-5400 unless otherwise specified herein.

3. 2.2 Service Conditions - The equipment shall operate satisfactorily under any " of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment.

3.2.3 . Standard Conditions - The following conditions shall be used as a basis to . establish normal performance requirements" and for making laboratory bench tests on the equip- . ment: "-

Temperature	Room ambient (30′ C ± 10° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input voltage	27.5 ± 0. 5V DC

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3. 2.4 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 8 pounds.

3. 2.5 Operational Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without the necessity for readjustment of' any controls which are inaccessible during flight.

3,2.6 Operating Life

.3. 2.6.1 Reliable Installed Operating Life - The equipment shall have a reliable installed operating life of at least 1000 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3. 2 6. 2 Total Installed Operating Life - The equipment shall have a minimum total operating life of six years with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such puts shall be specified by the manufacturer.

3. 2,7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3. 2. 8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. However, external cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3. 2.9 Console Controls - The equipment shall be satisfactory for mounting in a console arrangement when used in aircraft. This equipment shall meet the applicable requirements of Specification MIL-C-6781 and shall have a plastic lighting plate conforming to Specification MIL- P-7788. The lamps shall be connected in parallel with one side grounded to the chassis of the unit and the other side connected to a separate contact, in the electrical connector receptacle. of the unit, for connection to an external+ 28-volt input lead. The brilliance of the lamps will be controlled externally and the brilliance control shall not be part of this equipment. The arrangement of controls and all lettering and markings proposed for use on the front panel of the equipment shall be approved by the Bureau of Aeronautics prior to fabrication of any complete units. All dial settings shall be direct reading (e. g: 0475 yards)

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power source with characteristics as defined in Specification MIL-E-7894 having limits as specified hereinafter. The power required shall not exceed the specified amount:

(1) DC power 28-volt system

Operating voltage limits	24 to 30 volts
. Reduced performance voltage limits .	24 to 30 volts.
Current	15 amps into a 2.5 mh inductive load intermittent for firing an external load, plus 2 amps continuous

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3.2.11 Inn- Input power connections shall be in accordance with MS 25245.

3.2.12 Overload Protection- Overload protection for the equipment shall be provided in the equipment, All parts and circuits of the. equipment which. are likely to carry an overload due to anY failures or poor adjustments Shall have suitable protection devices or shall be . Proportioned to withstand such overload without permanent damage to the equipment. The use of . fuses and other protective devices shall be held to a minimum.

3.2.12.'1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2. 13 Standby Provisions - The equipment shall provide for a standby position wherein rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment. .

3.2.14 Nomenclature and Nameplates - Nomenclature assignment. and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307. .

3.2.15 Interchangeability y - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of Specification MIL-E-5400 conflict with the requirements for interchangeability y, the interchangeability y requirements shall govern.

3.2.16 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-MOO. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.17 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181. .

3.3 . Performance

3.3.1 Warm-Up Time - The warm-up time for the equipment shall not exceed one minute under standard and extreme service conditions.

3.3.2 Power Consumption - The total power consumption of the equipment shall not exceed 60 watts continuous and 510 watts during pulsing at 28-volts DC.

3.3.3 Accuracy - Under conditions of thermal equilibrium at an ambient room temperature of 30°C ± 10°C the accuracy for one pulse shall not be less than one part in 10,000, plus 1 millisecond possible variation from the mean spacing, for any one pulse.

-3.4 . Detail Requirements . . .

3.4.1 Intervalometer, Ordnance Release TD=239/A- The intervalometer shall be in accordance with Drawings 58A163D1, 58A163D1, 58A163D2, and 58A163J3 and in addition shall meet the following requirements.

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3.4.2 Form Factor and Mounting - The equipment shall conform to the requirements of 3. 2.9 and shall have overall dimensions of 5-3/4 inches wide by 7-7/8 inches long by 6-1/4 inches deep behind the panel.

3.4.3 Weight with Mounting - The installed weight of the equipment, with mounting, shall not exeed 8 pounds.

3.4.4 Input Pulse - The input pulse shall be + 24 to +30 V. D. C. into an input impedance of not less than 2 ohms for a minimum duration of 11 milliseconds. Negative transients of up to 150 volts and O. 1 120 second pulse duration at the input shall not initiate operation.

3.4.5 Output Pulse - The equipment shall provide output pulses of 0.0725 ± 0.010 seconds duration.

3.4.6 circuits - With the power switch in the "OFF" position, the equipment shall be completely shut off and by passed for normal selective release operation.

3.4.3 Range

GROUND SPEED -	1-600 knots in 1 knot intervals
Spacing	150-3200 yards in 25 yard intervals
	150-3200 feet in 25 foot intervals
Rounds	1-64 in 1 round intervals

"3. 5 - Workmanship, fabrication and manufacturing processes shall conform to Specification MIL-E-54

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4. QUALITY ASSurance PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) . Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(.1) Contractor's Demonstration Tests . . .

(2) Service Approval Tests "

4.2.1: . Contractor's Demonstration Tests - Contractor's demonstration tests . shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these "tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the" specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau pf Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4. 2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to . . repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for . . service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-E-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-E- 5422 and interference tests in accordance with Specification MIL-I-6181. The sampling . . and testing of plastic lighting plates shall be in accordance with-Specification MIL-P-7788.

4.2.4 Preprodutti on Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor. shall furnish test reports showing quantitative results for all acceptance tests. . Such reports shall be signed by an authorized representative of the contractor or laborator y as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment . accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3. 1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1."2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize, and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3. 1.3 Reliability Test - Each equipment shall be operated for a period of 30 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. , Only those adjustments will be permitted which can be made.. by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity -	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 30-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified 'limits of satisfactory performance is not necessarily required during the vibration - period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 30-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodially through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The. 30-hour period specified above may be adjusted to conform with the standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1 for each additional 200 or fraction thereof
Over 200	

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test

4.3.3.1. Special Tests Schedule - Special tests shall be conducted on the first and every fifth equipment that Successfully passes the sampling tests. However, additional equipments may be selected when required by the following:

(1) After an engineering change

(2) Whenever failure reports or other information indicate additional tests are necessary.

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minutes cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following:, (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
D. C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4..4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed. by the contractor shall be subject to approval by the Bureau of Aeronautics.

4,4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following

(1) Part type number

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(2) The circuit reference symbol-number

(3) The. part function

(4) Name of" the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of apart failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material correction for all failures" which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) . On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and -"wear" items. The "wear" . items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification on the contract. Specification MIL- T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shaIl be . submitted by the contractor until It has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the proocuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. . To obtain Bureau. of Aeronautics approval, . the contractor shall submit an Engineering Change Proposal (ECP). giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an 'Electronic Material Bulletin (EMB) to cover the approved change. ECP'S, EMC'S and EMB's shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged; packed and marked for the level of shipment specified in the contractor order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values- Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0. 5V DC. These data are to "be used in testing the-equipment at installation points for compliance with minimum acceptable standards of performance.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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